MIL-STD-1771
30 December 1981
Superseding
MIL-V-38352
20 January 1965

MILITARY STANDARD

VALUE ENGINEERING PROGRAM

REQUIREMENTS

FSC MISC

AMSC No. F0445

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20360

VALUE ENGINEERING PROGRAM REQUIREMENTS

MIL-STD-1771

1. This military standard is approved for use by all departments and agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: ASD/ENESS, Wright-Patterson AFB, OH 45433, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

FOREWORD

This standard has been designed to encourage the contractor to establish and maintain, throughout the life of the contract, a value engineering (VE) program.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

VALUE ENGINEERING PROGRAM REQUIREMENTS

1. SCOPE

1.1 General. This standard establishes the minimum requirements for a contractor's value engineering (VE) program and prescribes a minimum standard of contractor value engineering performance.

1.2 Applicability. This standard applies to a contractor's VE program when the contract contains a VE program requirement clause.

1.3 Objective. The objective of this standard is to define minimum requirements essential to the VE program.

2. REFERENCED DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

STANDARDS

MILITARY

DOD-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-481	Configuration Control - Engineering Changes, Deviations and Waivers (Short Form)
MIL-STD-499	Engineering Management

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Nongovernment documents. Direct cite of nongovernment documents will be determined and tailored to the immediate contract. The contractor is also encouraged to propose adoption of such nongovernment documents as may be developed to support a viable VE program.

3. DEFINITIONS AND ACRONYMS

3.1 Definitions

3.1.1 Value engineering/value analysis. An organized effort directed at analyzing the function of systems, equipment, facilities, services and supplies for the purpose of achieving the essential functions at the lowest overall cost

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

of ownership consistent with required performance, reliability, maintainability, interchangeability, product quality assurance and safety.

3.1.2 Value engineering program. The contractor's VE policies, procedures and organization, and planned activities as described in the VE plan directed toward application of the VE methodology as agreed to in the contract.

3.1.3 Essential characteristics. The customer's minimum essential, operational, functional, maintenance, safety, and reliability needs which must be fulfilled.

3.1.4 Value engineering change proposal. An engineering change proposal (ECP) submitted pursuant to a contract VE clause which proposes a contractual change that, if accepted, will provide the essential functions intended in the contract at a reduction in acquisition cost or the total system life cycle cost.

3.1.5 Life cycle costs. The total ownership cost to the Government over the estimated full life of the contracted supplies or services. It includes all costs associated with development, acquisition, operation, maintenance, support, and where applicable, disposal.

3.1.6 Value engineering study. The application of VE methodology to a specific subject or product area determined to have potential for cost reduction. A VE study is conducted by a team of people with skills tailored to the subject or product area proposed. Teams should normally possess engineering, production, logistic and purchasing talents.

3.2 Acronyms

ACO	Administrative contracting office
CDRL	Contract data requirements list
COTR	Contracting officer's technical representative
ECP	Engineering change proposal
GFE	Government furnished equipment
PCO	Principal contracting office
VE	Value engineering
VECP	Value engineering change proposal

4. GENERAL REQUIREMENTS

4.1 VE program

4.1.1 VE program plan. The contractor shall prepare a VE program plan and submit it in accordance with the contract data requirements list (CDRL). The plan shall be tailored to the requirements described in 4.1.1.1 through 4.1.1.7. The plan shall be definitized upon contract award and submitted as specified by the CDRL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

4.1.1.1 VE organization. The contractor's VE organization, policy and procedures shall be described. References to or samples of published organization charts, VE policies and procedures may be included in this section. Top management's VE decision structure shall be identified. The contractor point of contact in the VE organization for the contract shall be named. Successors may be nominated as required. The VE organization shall include individuals well versed in the principles, application and contractual aspects of value engineering. The contractor's VE point of contact shall be placed in an effective position in the corporate structure and shall have sufficient, well-defined responsibility, authority, and the organizational freedom to conduct the VE program.

4.1.1.2 VE training. The contractor's plans for training project and key decision-making personnel associated with this contract shall be described, including the type of VE training, manhours to be utilized, and disciplies to be represented (engineering, manufacturing, purchasing, estimating, etc.).

4.1.1.3 Work breakdown structures. The contractor's work breakdown structure common to engineering and production planning shall identify high cost areas and propose areas for potential VE study.

4.1.1.4 Specification review. The contractor shall review contract specification requirements and identify or recommend areas for potential VE study.

4.1.1.5 Vendor or subcontractor participation. The contractor shall provide for and encourage vendor or subcontractor participation in the VE program by flow down of applicable VE provisions from the prime.

4.1.1.6 VE program schedule. The contractor shall identify key VE events and milestone dates consistent with the overall program schedule.

4.1.1.7 VE funding. The contractor shall account for expenditure and allocation of the contract VE funds.

4.1.2 Communication. The contractor's VE manager shall maintain direct communication with the procuring agency VE personnel to facilitate the collection of information required by the approved VE program. VE studies shall identify alternate approaches that show potential for reducing life cycle costs. Should a contract change be required as a result, the principal contracting office (PCO) and the administrative contracting office (ACO) shall be notified of the potential value engineering change proposal (VECP), essential supporting information, and estimated completion date.

4.1.3 Program implementation. The contractor and government shall coordinate systematic procedures for conduct of the VE program and measuring its effectiveness. The VE program shall be integrated with the project management effort. MIL-STD-499 applies if implemented by the governing contract. The VE program actions shall be documented and maintained current during contract execution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

4.1.3.1 VE potential. Value engineering activity shall focus on but need not be limited to, the following areas:

a. Contract requirements

Technical
Support (including packaging, transportation and handling)
Data
Schedule

b. Hardware

Procured
Fabricated
Government furnished equipment (GFE)

c. Manufacturing

Procedures
Processes
Tooling
Test equipment and procedures

d. Installation

Equipment procured
Layout
Procedures

e. Operations

Policy
Layout
Procedures
Staffing

f. Maintenance

Repair policy
Repair procedure
Repair cycle or level
Test equipment

g. Facilities

h. Software

i. Testing

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

4.1.3.2 Cost goals. VE cost goals shall be established by the contractor within the project milestones through each phase for the various elements of the project when in production. The contractor shall allocate and adjust for realistic VE goals that aid in the identification of potential VE studies. Contractor management shall be kept informed of cost goal attainment status periodically.

4.1.3.3 Cost information. Related VE cost information sufficiently accurate for reliable decision-making shall be available to the government. The contractor's cost models setting forth major cost elements and their contribution to total cost will be used wherever possible to substantiate the VE program. This information, the cost goal information specified in 4.1.3.2, and the information provided in 4.1.1.3 shall be used for selecting VE study projects.

4.1.3.4 Contract requirements review. The contractor shall review the contractual requirements, identify those unique government requirements that contribute to poor value, including design, workmanship and production processes. The cost information developed in accordance with 4.1.3.3, shall serve as a baseline for this review. Technical interchange meetings between the contracting officer's technical representative (COTR), the ACO, and the contractor's VE representative for reviewing the need for the contract technical specifications are encouraged. At least one such review shall be held following each milestone to allow effective implementation of approved VECPs. This review shall address requirements which have the potential to minimize installation, operating, maintenance, and logistics costs throughout the useful life of the product.

4.1.3.5 Design reviews. VE personnel shall provide an input to both contractor internal and government/contractor design reviews. Value engineering recommendations, actions and results shall be documented and included in the design reviews. When formal design reviews are not conducted, a special VE review of design and support shall be held prior to release for production. This review procedure shall not be interpreted as being limited to construction, research and development, or other product oriented projects.

4.1.4 Subcontracts and acquisition. The contractor shall be encouraged to flow down appropriate VE arrangements in subcontracts in accordance with applicable acquisition regulations. The contractor shall assist, monitor and evaluate subcontractor performance in the area of value engineering.

4.1.5 Value engineering changes. When the results of a VE study indicate an alternate approach, relaxation or elimination of requirements would reduce life cycle costs, the contractor shall process the proposed change in accordance with 4.1.5.1 or 4.1.5.2. The estimated cost avoidance shall include and account for any costs required to implement the change.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

4.1.5.1 Government approval. A proposed VE change that requires a contractual change to implement shall be submitted to the principal contracting office (PCO) with a copy for the ACO for disposition in accordance with the contract terms. The VE proposal shall be clearly identified as a VECP and shall be prepared in accordance with DOD-STD-480 or MIL-STD-481 or as required by the contract.

4.1.5.2 Contractor approval. A proposed VE change which does not require a change to the contract to implement shall be evaluated and either approved or disapproved by the contractor. The contractor's normal procedures pertaining to such engineering changes apply and are subject to review by the government.

4.1.6 Relationship of value engineering/value analysis and design to cost. The relationship between value engineering/value analysis and related efforts such as design to cost shall be clarified when necessary to avoid potential duplication in effort and conflicts concerning incentives.

4.1.7 Documentation and reports

4.1.7.1 Documentation and access. The contractor shall maintain program files in sufficient detail to enable the PCO or his authorized representative to evaluate the quantity and quality of work performed pursuant to the VE program requirement. The documentation shall include VE study reports, cost goals, cost information, VE funding used by the project, reviews, and any other information required by the contract. Documentation pertaining to the contractor's required value engineering program shall be kept current and available for review by the contracting officer or his authorized representative.

4.1.7.2 Reports. The contractor may be required to prepare and submit sequential progress reports as specified in the CDRL schedule. Reports shall compare actual progress with planned and significant accomplishments to date.

4.1.7.3 Reviews. Progress of the VE effort shall be addressed at each program or design review.

4.2 Office of Management and Budget (OMB) clearance number. The OMB number (22-R0342) will be displayed prominently in the upper right corner of any forms associated with this military standard.

5. DETAILED REQUIREMENTS (Not applicable to this standard)

6. DATA ITEM DESCRIPTIONS (DIDs)

6.1 Associated DIDs. DIDs associated with this document are as follows:

DI-P-1600	Value Engineering Data Reports
DI-P-1602	Value Engineering Plan

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

Custodian activities:
Army - AR
Navy - SA
Air Force - 11

Review activities:
Army - AR
Navy - SH

Preparing activity:
Air Force - 11

Project No. MISC-0087

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 PREVIOUS EDITION WILL BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

ASD/ENESS
Wright-Patterson AFB, O 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-24T01:06Z
Check the source to verify that this is the current version before use.